

MGMT 590 Final Team Project

The final project is a research proposal-style project. Each team will choose one of two tracks: **LLM Alignment or LLM Evaluation**. The goal is to understand one important paper deeply, identify one concrete new research direction inspired by it, and explain why that direction is both practically important and methodologically feasible. Strong projects may later develop into an AI conference submission. A strong project should focus on one clear research question, rather than a broad survey topic. Evaluation emphasizes the quality of the idea, its clarity, and its feasibility.

Deliverables

1. In-class presentation

Each team will give a **≤10-minute presentation on April 27 or April 29**.

2. Final written report

The report is due on **May 1**. Please use the `neurips_2026.tex` template, available at <https://neurips.cc/Conferences/2026/CallForPapers>, and also on Brightspace. The report should be **≤ 8 pages** long, excluding references and appendix. It should include:

(a) Paper summary

Summarize the main idea, methodology, and key contribution of the chosen paper.

(b) New research direction

State one concrete new research question inspired by the chosen paper and explain why this new research direction matters in practice.

(c) Proposed solution

Describe one or more possible solutions to possibly solve this new research direction, and explain why the proposed approach is feasible and well motivated.

(d) Literature review

Review the relevant related work and explain the current state of the art.

(e) Preliminary evidence (optional but encouraged)

Preliminary numerical evidence is encouraged but not required. A toy simulation or exploratory analysis on a public dataset (e.g., prism data for alignment: <https://huggingface.co/datasets/HannahRoseKirk/prism-alignment>, or arena data: <https://huggingface.co/datasets/lmarena-ai/arena-human-preference-140k> for evaluation) is sufficient.

Evaluation Criteria

The project will be evaluated based on both the presentation and the written report.

Presentation (30%): content quality and technical understanding, and organization and presentation skills. The presentation is mainly an opportunity to communicate your idea clearly and receive feedback before the final report deadline.

Final report (70%): understanding of the chosen paper, originality and significance of the proposed new direction, quality of the literature review, clarity and feasibility of the proposed solution, and overall writing quality and organization.

Track 1: LLM Alignment

This track focuses on LLM alignment. The main ideas of *RLHF with PPO* and *DPO* have already been covered in class. Each team should:

1. use the course lectures as well as the following background references as the main background,
 - *Training Language Models to Follow Instructions with Human Feedback (RLHF-PPO paper)*
<https://arxiv.org/abs/2203.02155>
 - *Direct Preference Optimization: Your Language Model is Secretly a Reward Model (DPO)*
<https://arxiv.org/abs/2305.18290>
2. read the *RLHF survey paper* for broader context,
 - *RLHF Survey Paper*
<https://arxiv.org/abs/2604.02507>
3. and then choose **one advanced topic** from the following lists as the starting point for the final project. If you prefer to choose other advanced topics or papers, please discuss with the instructor in advance to get the approval first.
 - Heterogeneous / Pluralistic Feedback
 - *RLHF from Heterogeneous Feedback via Personalization and Preference Aggregation*
<https://arxiv.org/pdf/2405.00254>
 - *MixDPO: Modeling Preference Strength for Pluralistic Alignment*
<https://arxiv.org/abs/2601.06180>
 - Uncertainty Quantification
 - *Uncertainty Quantification for Large Language Model Reward Learning under Heterogeneous Human Feedback*
<https://arxiv.org/abs/2512.03208>
 - *RewardUQ: A Unified Framework for Uncertainty-Aware Reward Models*
<https://arxiv.org/abs/2602.24040>
 - Fairness
 - *Fairness Aware Reward Optimization*
<https://arxiv.org/abs/2602.07799>
 - *Intra-Fairness Dynamics: The Bias Spillover Effect in Targeted LLM Alignment*
<https://arxiv.org/abs/2602.16438>
 - Privacy
 - *Privacy-Preserving Instructions for Aligning Large Language Models*
<https://arxiv.org/abs/2402.13659>
 - *Privacy-Preserving Reinforcement Learning from Human Feedback via Decoupled Reward Modeling*
<https://arxiv.org/abs/2603.22563>
 - Causal Inference
 - *Preference Learning for AI Alignment: a Causal Perspective*
<https://openreview.net/pdf?id=iuD649wPAw>
 - *NextQuill: Causal Preference Modeling for Enhancing LLM Personalization*
<https://arxiv.org/pdf/2506.02368v3>

Track 2: LLM Evaluation

This track focuses on *LLM evaluation*, especially in open-ended settings where simple task accuracy is not enough. Core topics include Arena-style human pairwise evaluation and LLM-as-a-Judge have been discussed in the class. Each team should:

1. use the course lectures as well as the following background references as the main background,
 - *Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference*
<https://arxiv.org/abs/2403.04132>
 - *JudgeLM: Fine-tuned Large Language Models are Scalable Judges*
<https://arxiv.org/abs/2310.17631>
 - *A Survey on LLM-as-a-Judge*
<https://arxiv.org/abs/2411.15594>
2. and then choose **one advanced topic** from the following lists as the starting point for the final project. If you prefer to choose other advanced topics or papers, please discuss with the instructor in advance to get approval first.
 - Arena-Style Human Evaluation and Ranking
 - *Prompt-to-Leaderboard*
<https://arxiv.org/abs/2502.14855>
 - *Fewer Battles, More Gain: An Information-Efficient Framework for Arena-based LLM Evaluation*
<https://openreview.net/forum?id=XUVqFRp9oi>
 - Uncertainty Quantification and Reliability of Rankings
 - *A Judge-Aware Ranking Framework for Evaluating Large Language Models*
<https://arxiv.org/abs/2601.21817>
 - *Quantifying Uncertainty in LLM Benchmark Scores*
<https://arxiv.org/abs/2410.03492>
 - Critiques and Failure Modes of Arena-Style Leaderboards
 - *The Leaderboard Illusion*
<https://arxiv.org/abs/2504.20879>
 - *Dropping Just a Handful of Preferences Can Change Top Leaderboard Rankings*
<https://arxiv.org/abs/2508.11847>
 - LLM-as-a-Judge: Bias and Inference
 - *Judging the Judges: A Systematic Study of Position Bias in LLM-as-a-Judge*
<https://arxiv.org/abs/2406.07791>
 - *Efficient Inference for Noisy LLM-as-a-Judge Evaluation*
<https://arxiv.org/pdf/2601.05420>
 - Low-rank Evaluation
 - *Rich Insights from Cheap Signals: Efficient Evaluations via Tensor Factorization*
<https://arxiv.org/abs/2603.02029>
 - *LLM Evaluation as Tensor Completion: Low Rank Structure and Semiparametric Efficiency*
<https://arxiv.org/abs/2604.05460>